DSA SHORT-MARKS CRASH SHEET

1-mark + 2-mark questions from Units 1-3 | Based on the uploaded class materials and Mam's listed topics

Use this as your fast-revision sheet. 1-mark answers are one clean line. 2-mark answers are kept to two compact points.

UNIT 1 - LINKED LISTS

2-MARK QUESTIONS

Q1. What is a linked list?

A sequence of nodes in which each node stores data and a link/pointer to the next node.

Q2. What is a singly linked list?

Each node has data and one next pointer; the last node points to NULL.

Q3. What is a doubly linked list?

Each node has two pointers, prev and next, so traversal is possible in both directions.

Q4. What is a circular linked list?

A linked list in which the last node points back to the first node, forming a circle.

Q5. Difference between array and linked list?

Array: contiguous memory and fixed-size style access. Linked list: nodes linked by pointers and dynamic in size.

Q6. Difference between singly and doubly linked list?

SLL has one next pointer and forward traversal. DLL has prev + next pointers and supports both directions.

Q7. State two advantages of linked lists.

Dynamic size; insertion/deletion can be done by changing links without shifting elements.

Q8. What is traversal in a linked list?

Visiting each node from the head and moving through next links until NULL (or back to head for circular lists).

Q9. What is the last node of an SLL identified by?

Its next pointer is NULL.

Q10. What is the time complexity of searching an SLL?

O(n), because nodes are checked sequentially.

1-MARK RAPID FIRE

Question	Answer
1. What is the first node of a linked list called?	Head.
2. What does NULL indicate in an SLL?	End of the list.
3. How many pointer fields does an SLL node have?	One.
4. How many pointer fields does a DLL node have?	Two: prev and next.
5. Which linked list supports backward traversal?	Doubly linked list.
6. In a circular SLL, what does the last node point to?	Head.
7. What is insertion at the beginning of an SLL?	Make the new node point to head, then update head to the new node.
8. What is the complexity of insertion at the head of an SLL?	O(1).
9. What is the complexity of traversal of an SLL?	O(n).
10. What is the extra field in a DLL compared with SLL?	prev pointer.

UNIT 2 - STACK, QUEUE, INFIX-POSTFIX

2-MARK QUESTIONS

Q1. What is a stack?

A linear data structure that follows LIFO: Last In, First Out.

Q2. What are the basic stack operations?

Push inserts at the top; Pop removes the top element. Peek reads the top without removing it.

Q3. What is stack overflow and underflow?

Overflow occurs when pushing into a full stack; underflow occurs when popping from an empty stack.

Q4. What is an array representation of stack?

A fixed-size array stores elements and a variable top stores the index of the top element.

Q5. What is a queue?

A linear data structure that follows FIFO: First In, First Out.

Q6. What are the basic queue operations?

Enqueue inserts at rear; Dequeue removes from front.

Q7. What are front and rear in a queue?

Front points to the first element; rear points to the last element.

Q8. What is infix notation?

An expression in which the operator is written between operands, e.g. A+B.

Q9. What is postfix notation?

An expression in which the operator follows its operands, e.g. AB+.

Q10. State operator precedence used in infix-postfix conversion.

^ has highest precedence, then * and /, then + and -.

Q11. What is the role of stack in infix-to-postfix conversion?

It temporarily stores operators and parentheses so precedence and associativity are handled correctly.

Q12. What happens when) is read during conversion?

Pop operators to postfix until (is found; then discard it.

1-MARK RAPID FIRE

Question	Answer
1. LIFO stands for?	Last In, First Out.
2. FIFO stands for?	First In, First Out.
3. Which variable identifies the top of an array stack?	top.
4. Initial value of top in the notes?	-1.
5. What is push?	Insert an element into the stack.
6. What is pop?	Remove the top element from the stack.
7. What is peek?	Return the top element without removing it.
8. What are the queue end names?	front and rear.
9. Where is a queue element inserted?	At the rear.
10. Where is a queue element deleted?	From the front.
11. Which notation writes operator after operands?	Postfix.
12. Which operator has highest precedence in the notes?	^.
13. What is the associativity of ^ in the notes?	Right to left.
14. What happens to an operand in infix-postfix conversion?	It is appended directly to the postfix expression.
15. What happens when an opening parenthesis is read?	Push it onto the operator stack.
16. What is the example A*(B+C)/D in postfix?	ABC+*D/.

UNIT 3 - TREES, BST, HEAP, THREADED TREE

2-MARK QUESTIONS

Q1. What is a tree?

A non-linear data structure of nodes connected by edges, organized hierarchically.

Q2. Differentiate height and depth.

Depth is the number of edges from root to a node. Height is the longest number of edges from a node to a leaf.

Q3. What is a binary tree?

A tree in which each node has at most two children: left and right.

Q4. What is a full binary tree?

Every internal node has exactly two children; no node has only one child.

Q5. What is a perfect binary tree?

All internal nodes have two children and all leaves are at the same level.

Q6. What is a complete binary tree?

All levels are full except possibly the last, and the last level is filled from left to right.

Q7. What is a BST?

A binary tree where left-subtree values are smaller than the root and right-subtree values are greater (as framed in the notes).

Q8. What are the three BST deletion cases?

Delete a leaf; delete a node with one child; delete a node with two children.

Q9. What is a binary heap?

A complete binary tree used for priority queues, satisfying min-heap or max-heap order.

Q10. Differentiate min heap and max heap.

Min heap: parent <= children. Max heap: parent >= children.

Q11. What is a threaded binary tree?

A binary tree that uses otherwise-null links as threads to help traversal.

Q12. What is the maximum number of nodes at level L?

2^L.

Q13. What is the maximum number of nodes in a binary tree of height H?

2^(H+1) - 1.

Q14. What is the number of edges in a non-empty tree with N nodes?

N - 1.

1-MARK RAPID FIRE

Question	Answer
1. What is the topmost node of a tree called?	Root.
2. What is a node with no children called?	Leaf node.
3. What are nodes with the same parent called?	Siblings.
4. What is the number of children of a node called?	Degree of the node.
5. What is the root depth in the notes?	0.
6. What is the height of a leaf in the notes?	0.
7. What is a parent node?	A node having one or more children.
8. What is an internal node?	A node with at least one child.
9. Which traversal is Root-Left-Right?	Preorder.
10. Which traversal is Left-Root-Right?	Inorder.
11. Which traversal is Left-Right-Root?	Postorder.
12. Which traversal gives BST nodes in increasing order?	Inorder.
13. How many children can a binary-tree node have at most?	Two.

Question	Answer
14. What is a left-skewed tree?	A skewed tree whose nodes are dominated by left children.
15. What is a right-skewed tree?	A skewed tree whose nodes are dominated by right children.
16. What is the balance-factor range for an AVL node in the notes?	-1, 0, or +1.
17. What does BST search do if target is smaller than root?	Move to the left subtree.
18. What does BST search do if target is larger than root?	Move to the right subtree.
19. What is the parent-child order in a max heap?	Parent is greater than or equal to children.
20. What is the parent-child order in a min heap?	Parent is less than or equal to children.
21. What is heapify/sift-up used for?	Restoring heap order by comparing and swapping with a parent/child as needed.
22. Which tree type is used for priority queues?	Heap tree.
23. What is a general tree?	A tree where a node may have any number of children.
24. What is the maximum number of edges in a tree with N nodes?	N - 1.

FAST EXAM MEMORY

Topic	Memory hook
SLL	1 pointer -> next -> last node NULL
DLL	2 pointers -> prev + next
CLL	last node -> head
Stack	LIFO | top | push/pop
Queue	FIFO | front/rear | enqueue/dequeue
Infix -> Postfix	operand direct | operators in stack | precedence
Tree	root | parent | child | leaf | degree | level
Height vs Depth	height down to leaf | depth down from root
BST	left < root < right
Heap	complete tree | min: parent <= child | max: parent >= child

Source note: Questions and compact answers are derived from the uploaded Unit 1, Unit 2, and Unit 3 course materials. Mam's listed topics were used to prioritize coverage; this is a revision bank, not a guarantee of exact exam wording.